

MIDELT, Morocco

WMO: 601950

Lat: 32.683N

Lon: 4.733W

Elev: 1515

StdP: 84.40

Time Zone: 1.00 (E01)

Period: 90-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-0.7	-11.2	1.7	8.5	-9.1	2.1	8.0	12.8	10.2	10.4	5.8	2.3	260	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	34.4	17.3	33.2	16.9	32.0	16.6	19.6	30.9	18.7	29.8	17.9	28.7	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	13.6	23.5	14.8	12.7	23.0	14.0	12.0	22.7	63.0	31.2	59.7	29.8	56.9	28.7	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.2	7.5	-4.3	36.7	2.0	1.3	-5.7	37.6	-6.8	38.3	-7.9	39.1	-9.4	40.0
			-4.9	21.1	1.9	1.9	-6.3	22.4	-7.4	23.5	-8.5	24.6	-9.9	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	6.4	8.2	10.8	13.1	16.7	21.6	25.4	24.8	20.4	15.8	10.8	7.8
	DBStd	7.16	3.16	3.34	3.69	3.38	4.11	3.41	2.48	2.23	3.09	2.96	3.45	2.87
	HDD10.0	346	117	67	35	12	2	0	0	0	0	3	32	78
	HDD18.3	1788	369	284	233	159	80	11	0	0	14	84	226	327
	CDD10.0	2241	6	17	60	103	211	347	477	458	313	183	56	10
	CDD18.3	642	0	0	0	1	31	108	218	199	77	7	0	0
	CDH23.3	6458	0	0	7	30	311	1077	2383	1974	627	49	0	0
	CDH26.7	2724	0	0	0	1	77	423	1151	885	186	1	0	0
Wind	WSAvg	2.4	2.3	2.4	3.0	3.2	2.6	2.3	2.1	2.0	2.1	2.2	2.5	2.3
Precipitation	PrecAvg	191	13	14	18	16	21	12	7	13	21	24	20	14
	PrecMax	328	41	63	36	44	73	51	32	45	61	115	121	45
	PrecMin	121	0	0	4	0	1	0	0	0	1	1	0	1
	PrecStd	49	12	14	8	13	20	13	7	12	15	25	24	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	19.8	24.1	26.0	31.0	34.3	36.2	35.2	32.8	26.6	22.1	18.1
		MCWB	8.4	8.9	11.0	12.8	15.4	16.1	17.8	17.8	16.8	14.3	11.8	7.9
	2%	DB	15.5	18.1	21.6	24.0	28.9	32.6	34.9	33.8	30.4	24.9	20.3	16.0
		MCWB	7.3	8.3	10.2	11.9	14.9	15.5	17.4	17.4	16.1	14.0	11.1	7.9
	5%	DB	13.6	16.5	19.8	22.1	26.9	31.0	33.7	32.7	28.9	23.3	18.8	14.3
		MCWB	6.4	7.4	9.6	11.1	14.4	15.5	17.2	17.0	15.6	13.5	10.4	6.9
	10%	DB	11.8	14.7	17.9	20.2	24.8	29.2	32.5	31.4	27.1	21.8	16.9	12.7
		MCWB	5.5	6.9	8.8	10.4	13.5	15.3	16.9	16.8	15.3	13.0	9.3	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	11.2	12.6	15.0	17.0	19.5	20.7	21.3	18.8	17.1	14.2	10.9
		MCDB	15.9	16.9	21.1	23.8	27.5	30.7	32.0	31.4	28.2	23.3	19.7	14.3
	2%	WB	8.5	9.7	11.0	13.2	15.7	18.1	19.4	19.8	17.7	15.7	12.3	9.3
		MCDB	13.6	15.4	19.4	21.0	26.0	28.2	31.9	30.4	26.8	22.6	18.4	13.2
	5%	WB	7.2	8.4	10.1	12.0	15.0	16.9	18.5	18.9	16.9	14.7	11.1	8.3
		MCDB	12.3	14.5	18.0	19.8	25.4	26.6	30.8	29.4	25.9	20.9	17.3	12.2
	10%	WB	6.1	7.2	9.3	11.0	14.1	16.0	17.7	18.0	16.2	13.8	10.0	7.3
		MCDB	11.2	13.6	16.6	18.7	23.5	25.6	29.8	28.4	24.3	19.7	15.3	11.4
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.5	11.1	11.8	12.2	13.5	14.0	13.5	12.1	10.7	10.1	9.4
		MCDBR	11.9	13.3	13.9	14.8	14.5	15.5	15.2	15.1	14.2	12.7	12.3	11.7
		MCWBR	6.6	6.8	6.3	6.4	5.5	4.7	4.9	4.9	5.1	5.6	6.2	6.3
	5% WB	MCDBR	10.9	11.8	12.5	13.7	13.8	14.2	14.3	13.9	13.0	11.2	11.2	9.8
		MCWBR	6.6	6.7	6.0	6.5	5.8	5.4	5.7	5.6	5.4	5.9	6.4	6.1
Clear-Sky Solar Irradiance	taub	0.242	0.274	0.322	0.345	0.382	0.417	0.478	0.446	0.381	0.339	0.280	0.251	
	taud	2.645	2.521	2.382	2.334	2.243	2.132	1.940	2.052	2.271	2.402	2.563	2.619	
	Ebn at Noon	1007	992	957	943	905	868	814	836	882	900	944	975	
	Edh at Noon	73	92	115	126	140	156	188	166	127	103	79	71	
All-Sky Solar Radiation	RadAvg	3.51	4.43	5.59	6.61	7.22	7.80	7.42	6.70	5.78	4.56	3.61	3.22	
	RadStd	0.29	0.33	0.33	0.43	0.37	0.34	0.28	0.27	0.33	0.25	0.28	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.39	N/A	N/A	+1.28	N/A	N/A	N/A	N/A	+195	+191

Nomenclature: See separate page